A SIMPLE NON-ABSOLUTELY-SIMPLE ALGEBRA IS R-UNIVERSAL EXACTLY WHEN IT IS THE RESTRICTION OF SCALARS OF A STRONGLY INNER ALGEBRA OVER AN IMAGINARY QUADRATIC EXTENSION

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Allowed standard prerequisites (editorial).

- The description of a simple non-absolutely-simple algebra as the restriction of scalars of an absolutely simple algebra over a finite extension. Used at the start of the counted proof (source lines 857-858), attributed by the author to section 3.1.2 of the cited paper Tits-Classification. Supplies the shape of the algebra that the criterion is stated in terms of.
- The relation between the beta invariants of an algebra and those of its restriction of scalars. Used inside the counted proof where the author compares the invariant of the restricted weight with the invariant of the original one (source lines 859-877). Transfers the vanishing condition for the beta invariant across the restriction of scalars.

Source reference. Original definition 1.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 135–135. Navigation pointer only: the author's definition of the beta invariant, used by the counted proof in its notation.

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Source reference. Original lemma 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 416–416. Navigation pointer only: the counted proof invokes the author's equivalence for R-universality; that lemma has its own card in this lane and its content is declared as a named standard prerequisite.

Source reference. Original definition 4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 432–432. Navigation pointer only: the labelled item of that equivalence used by the counted proof.

Source reference. Original lemma 2.3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 471–471. Navigation pointer only: the author's lemma on the minimal inner extension; it has its own card in this lane and its content is declared as a named standard prerequisite.

Proposition 0.1. *Let $\mathfrak{g}$ be a simple Lie algebra over $\mathbb{Q}$ that is not absolutely simple. Then $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if $\mathfrak{g} = \mathbf{R}_{L/\mathbb{Q}}\,\mathfrak{g}'$ is the restriction of scalars of a strongly inner, absolutely simple Lie algebra $\mathfrak{g}'$ over an imaginary quadratic extension L of $\mathbb{Q}$.*

Proof. ($\Rightarrow$) We have $\mathfrak{g} = \mathbf{R}_{L/\mathbb{Q}}\,\mathfrak{g}'$, for some absolutely simple Lie algebra $\mathfrak{g}'$ over some finite extension L of $\mathbb{Q}$ [10, § 3.1.2]. From Lemma 2.3, we see that if $\mathfrak{g}$ is $\mathbb{R}$-universal, then the extension L must be imaginary quadratic.

Let λ' be a nonzero dominant weight of $\mathfrak{g}'$, and let $\lambda = \mathbf{R}_{L/\mathbb{Q}}\,\lambda'$ be the corresponding weight of $\mathfrak{g}$. Then λ is not fixed by the $*$-action, so we have $D_{\mathfrak{g},F}(\lambda) = D_{\mathfrak{g},L}(\lambda) = D_{\mathfrak{g}',L}(\lambda)$. (The final equality is because $\mathfrak{g} = \mathbf{R}_{L/\mathbb{Q}}\,\mathfrak{g}'$.) On the other hand, since all Lie algebras are split over $\mathbb{C}$, we have $D_{\mathfrak{g},\mathbb{R}}(\lambda) = D_{\mathfrak{g},\mathbb{C}}(\lambda) = \mathbb{C}$. Hence, we see from Lemma 2.1(4) that if $\mathfrak{g}$ is $\mathbb{R}$-universal, then $D_{\mathfrak{g}',L}(\lambda) = L$. Since this is true for every dominant weight λ of $\mathfrak{g}'$, we conclude that $\mathfrak{g}'$ is strongly inner.

($\Leftarrow$) Note that, since $\mathfrak{g}'$ is strongly inner, the simply connected group $\mathbf{G}'$ corresponding to $\mathfrak{g}'$ may be written as $\mathbf{G}' = {}^{\zeta}\mathbf{G}^s$, where $\mathbf{G}^s$ is split (and simply connected) and $\zeta \in H^1(L; \mathbf{G}^s)$. By Restriction of Scalars, then $\mathbf{G} = {}^{\xi}\mathbf{G}^q$, where $\mathbf{G}^q = \mathbf{R}_{L/\mathbb{Q}}\,\mathbf{G}^s$ is quasi-split and $\xi \in H^1(\mathbb{Q}; \mathbf{G}^q)$. Then, in the notation of Definition 1.1, we have $\delta_*[\xi] = 0$, so $\beta_{\mathfrak{g},F}(\lambda) = 0$ for every $*$-invariant weight λ. It is therefore easy to see that $\mathfrak{g}$ is $\mathbb{R}$-universal. □

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